

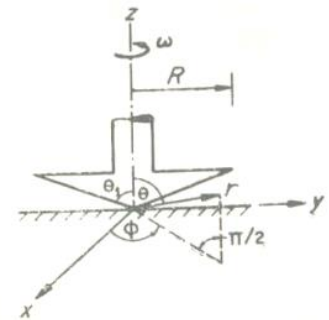
1. A sealed journal bearing is formed from concentric cylinders. The inner and outer radii are 25 and 26 mm, the journal length is 100mm, and it turns at 2800rpm. The gap is filled with oil in laminar motion. The torque needed to turn the journal bearing is 0.2Nm. Calculate the viscosity of the oil. Will the torque increase or decrease with time? Why?

2. A viscous shear pump is made from a stationary housing with a close-fitting rotating drum inside. **The clearance is small compared to the diameter of the drum.** Fluid is dragged around the annulus by viscous forces. Evaluate the performance characteristics of the shear pump (pressure differential, input power and efficiency) as functions of volumetric flow rate. Assume that the depth normal to the diagram is b . Prove that for flow in between the two plates forming a thin channel (width large compared to the gap), the flow regime is independent of the gap height for a fixed flow rate.

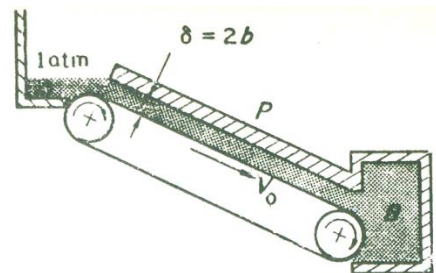
3. A cone and plate viscometer consists of a cone rotating at a constant angular velocity ω over a stationary plate. Liquid fills the gap between the cone and plate and the torque required to keep the plate stationary is measured through the deflection of a tensional spring. Simplify the Navier-Stokes equation assuming that flow is laminar and tangential (i.e., only $v_\phi \neq 0$), and that the inertial and gravity terms are negligible. Assuming $v_\phi = rf(\theta)$, show that $f(\theta)$ must satisfy

$$f'' + f' \cot \theta + f \left(2 - \frac{1}{\sin^2 \theta} \right) = 0$$

If the cone angle $[\pi/2 - \theta_1]$ is small, the velocity V_ϕ may be fairly well approximated as linear in the vertical direction. Compute the torque on the lower plate in terms of μ , R , ω , and θ_1 . If $R=5\text{cm}$, $\theta_1=89^\circ$, torque= $5 \times 10^{-3} \text{ Nm}$, $\omega = 2\text{rpm}$, determine the viscosity of the liquid.



4. A belt of width W moves at velocity V_o as shown in the figure. A liquid fills the gap (of thickness $\delta = 2b$) between the belt and an upper plate. The upper plate is dragged down to the tank B with a velocity equal to V_o . After some time, the tank B gets filled up (without any leakage and thus stopping the flow) and steady state conditions prevail.



a) Obtain the velocity profile and the pressure gradient for this condition. Please take the origin of the coordinate system to be on the belt with the direction of the belt movement as z and the height of the film as y . The effect of gravity may not be neglected. θ is the angle of inclination with respect to the horizontal.

b) A weigh-bridge is designed based on the system just described. The object to be weighed is placed on the horizontal platform P and the belt is started at velocity V_o and steady state is reached. The film thickness δ is directly calibrated to give the load (W). Evaluate the relation between the load and the film thickness (δ) in terms of V_o , properties (μ and ρ) and system geometry (L , $2b$, θ)

5. The space between two very long concentric cylinders is filled with water. The inner cylinder is stationary while the outer cylinder is pulled along axially with a velocity V_o .

a) Starting with the relevant governing equation and boundary conditions evaluate the velocity profile and the stress at the surface of the inner cylinder.

b) What is the volumetric flow rate of water at any axial location?

